

Introduction to Computer Networks (CN) — Detailed Descriptive Notes

What is a Computer Network?

Technical Definition

Computer Networking

A computer network is a system in which multiple computers and devices are connected together through communication links so that they can exchange data, share resources, and communicate using standard protocols.

Simple Understanding

Suppose you have two computers. If both computers are connected using cables or wireless technology and they can exchange files, messages, or internet access, then they form a computer network.

Today almost everything works because of computer networks:

- Internet
- WiFi
- Mobile communication
- Banking systems
- Cloud computing
- Online gaming
- Social media

All depend on networking.

Why Computer Networks are Important?

Computer networks are important because modern communication and resource sharing are impossible without them.

Earlier, if one computer had a printer or internet connection, other computers could not use it. Networking solved this problem by allowing devices to share resources and communicate efficiently.

Main Purposes of Networking

1. Resource Sharing

One of the biggest purposes of networking is sharing resources.

Resources include:

- Printers
- Files
- Internet connection
- Storage devices
- Software

For example:

In an office, instead of buying 50 printers for 50 employees, one network printer can be shared among all employees.

This reduces:

- Hardware cost
- Maintenance cost
- Space usage

2. Communication

Networking allows devices and users to communicate instantly.

Examples:

- Emails
- WhatsApp
- Video conferencing
- Voice calls
- Online meetings

Applications like:

- Zoom
- Google Meet
- Microsoft Teams

all use computer networking.

Without networking, real-time communication across countries would not exist.

3. Data Sharing

Networks allow users to access and transfer data from one system to another.

For example:

A company stores data on a server. Employees can access that data from their own computers through the network.

This makes collaboration easy.

4. Centralized Management

In large organizations, data and security are managed centrally.

This means:

- Admin controls all systems
- Backup can be managed centrally
- Security policies applied easily

Example:

College labs controlled by one server.

5. Cost Reduction

Networking reduces overall cost because:

- Resources are shared
 - Maintenance becomes easier
 - Data management centralized
-

Components of Computer Network

A computer network requires several basic components to work properly.

1. Sender

The sender is the device that sends data.

Examples:

- Laptop sending email
- Mobile sending WhatsApp message

The sender converts information into signals before transmission.

2. Receiver

The receiver is the device that receives the transmitted data.

Examples:

- Server receiving request
- Friend receiving your message

The receiver converts signals back into understandable information.

3. Transmission Medium

Transmission medium is the path through which data travels.

It can be:

- Wired
- Wireless

Examples:

- Fiber optic cable
- Ethernet cable
- WiFi
- Bluetooth

Without transmission media, communication cannot occur.

4. Protocol

Technical Definition

Communication Protocol

A protocol is a set of rules that defines how devices communicate in a network.

Protocols determine:

- How data is sent
- How errors are handled
- How devices identify each other

Example:

TCP/IP protocol suite.

Protocols are important because devices made by different companies must communicate correctly.

5. Network Devices

Network devices help establish communication between systems.

Examples:

- Hub
- Switch
- Router
- Modem

These devices manage traffic and connectivity.

Types of Computer Networks

Networks are classified based on geographical coverage area.

1. PAN (Personal Area Network)

PAN is the smallest type of network.

It covers only a few meters around a person.

Examples:

- Bluetooth connection
- Mobile hotspot
- Smartwatch connected to phone

PAN is used for personal device communication.

2. LAN (Local Area Network)

LAN covers a small area like:

- Home
- School
- Office
- College lab

LAN provides:

- High speed
- Low delay
- Better security

Example:

Computers connected in a school computer lab.

Most LANs use:

- Ethernet
 - WiFi
-

3. MAN (Metropolitan Area Network)

MAN covers a city or metropolitan area.

It is larger than LAN but smaller than WAN.

Examples:

- City-wide cable TV network
 - Networks connecting multiple offices across city
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4. WAN (Wide Area Network)

WAN covers very large geographical areas like:

- Countries
- Continents
- Entire world

The biggest example of WAN is:
Internet.

WAN uses:

- Satellites
 - Fiber optics
 - Telephone lines
-

Network Topology

Technical Definition

Network Topology

Network topology refers to the physical or logical arrangement of devices and communication links in a network.

Topology determines:

- How devices connect
- How data flows
- Network performance

Types of Topology

1. Bus Topology

In bus topology, all devices share one common cable.

When one device sends data, it travels through entire cable.

Advantages:

- Cheap
- Easy installation

Disadvantages:

- Cable failure stops entire network
 - Difficult troubleshooting
-

2. Star Topology

In star topology, all devices connect to one central device like switch or hub.

Most modern networks use star topology.

Advantages:

- Easy management
- Better performance
- Failure of one cable affects only one device

Disadvantage:

- Central device failure affects entire network
-

3. Ring Topology

In ring topology, devices form a circular connection.

Data travels in one direction.

Advantages:

- Organized communication

Disadvantages:

- One device failure can affect entire network
-

4. Mesh Topology

In mesh topology, every device connects to every other device.

Advantages:

- Highly reliable
- No single point failure

Disadvantages:

- Very expensive
- Complex setup

Used in:

- Military networks
 - Critical systems
-

Transmission Media

Transmission media carry data signals between devices.

Guided Media (Wired)

Guided media uses physical cables.

Twisted Pair Cable

Most common cable in LAN.

Cheap and easy installation.

Coaxial Cable

Better shielding against noise.

Used in cable TV systems.

Optical Fiber

Uses light signals.

Provides:

- Very high speed
- Long distance communication

- High security

Used in:

- Internet backbone
 - High-speed networks
-

Unguided Media (Wireless)

Wireless communication without physical cables.

Examples:

- WiFi
- Bluetooth
- Satellite communication
- Radio waves

Wireless media provides mobility and flexibility.

IP Address

Technical Definition

IP Address

An IP address is a unique logical address assigned to devices for identification and communication on a network.

Example:

192.168.1.1

Types of IP Address

IPv4

32-bit address.

Limited address space.

IPv6

128-bit address.

Created because IPv4 addresses are running out.

Provides enormous address capacity.

MAC Address

Technical Definition

MAC Address

MAC address is a unique physical address assigned to the network interface card of a device.

Unlike IP:

- MAC is hardware-based
- Usually permanent

Difference Between IP and MAC

IP address identifies device logically across networks.

MAC address identifies device physically inside local network.

OSI Model Introduction

Technical Definition

OSI Model

OSI model is a 7-layer conceptual framework used to understand how network communication occurs.

Each layer performs specific tasks.

OSI Layers

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer

Importance of OSI Model

OSI model helps:

- Understand networking
 - Troubleshoot problems
 - Standardize communication
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TCP/IP Model

Technical Definition

TCP/IP

TCP/IP is the practical networking model used on the Internet.

All internet communication depends on TCP/IP.

Real Life Example of Networking

When you open Google:

1. Browser sends DNS request
2. DNS returns IP address
3. TCP connection established
4. HTTPS secure communication starts
5. Data packets travel across routers and switches
6. Webpage displayed on screen

This entire process happens within seconds because of networking technologies.

Important Interview Questions

1. What is computer network?
2. Why networking needed?
3. Difference between LAN and WAN?
4. What is topology?
5. Difference between IP and MAC?
6. What is protocol?
7. What is TCP/IP?
8. What is OSI model?
9. Difference between wired and wireless communication?
10. What happens when you type google.com?

Voice chat ended

Network Devices — Detailed Descriptive Notes

Introduction to Network Devices

Technical Definition

Network Device

Network devices are hardware components used to connect computers and other electronic devices in a network so they can communicate, exchange data, share resources, and access services like the Internet.

These devices control:

- Data transmission
- Signal forwarding
- Traffic management
- Network security
- Communication between networks

Without network devices, modern networking and Internet communication would not be possible.

Why Network Devices are Important?

Imagine thousands or millions of computers connected worldwide. There must be some intelligent hardware that:

- Decides where data should go
- Connects devices
- Manages communication
- Prevents collisions
- Provides security

These tasks are performed by network devices.

Main Functions of Network Devices

Network devices perform many important tasks such as:

- Connecting devices
- Forwarding data
- Amplifying weak signals

- Routing packets
- Providing wireless communication
- Protecting networks from attacks
- Managing network traffic

Different devices perform different specialized functions.

Types of Network Devices

Major networking devices include:

1. Hub
 2. Switch
 3. Router
 4. Bridge
 5. Repeater
 6. Gateway
 7. Modem
 8. Access Point
 9. Firewall
 10. NIC (Network Interface Card)
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1. HUB

Technical Definition

Network Hub

Hub is a basic networking device that connects multiple computers in a LAN and broadcasts incoming data to all connected devices.

Hub works on:

- Physical Layer (Layer 1)
-

Basic Working of Hub

Hub is a very simple device.

When one computer sends data:

- Hub receives signal
- Copies signal
- Sends it to ALL connected devices

Hub does NOT know:

- Which device is sender
- Which device is receiver

It simply broadcasts everything.

Example

Suppose:

- PC1 wants to send data to PC3

Hub sends that data to:

- PC2
- PC3
- PC4
- Every connected device

Only intended receiver accepts it.

Others ignore it.

Why Hub is Inefficient?

Because every device receives every message.

This creates:

- Unnecessary traffic
 - Slow communication
 - Collisions
-

Collision

Technical Definition

Collision

Collision occurs when multiple devices send data at the same time and signals overlap.

Hub networks suffer from many collisions.

Collision Domain

Entire hub network forms one collision domain.

Meaning:

All devices compete for same communication medium.

More devices → More collisions.

Characteristics of Hub

Feature	Description
Layer	Physical Layer
Intelligence	No
Addressing	None
Communication	Broadcast
Speed	Slow
Duplex	Half duplex

Types of Hub

1. Passive Hub

Simply forwards signals.

No amplification.

2. Active Hub

Amplifies weak signals before forwarding.

3. Intelligent Hub

Can monitor network traffic.

Provides basic management.

Advantages of Hub

- Cheap
 - Simple
 - Easy installation
-

Disadvantages of Hub

- Very slow
 - High collisions
 - Poor security
 - Wastes bandwidth
-

Real Life Analogy

Teacher shouting in classroom.

Everyone hears message even if intended for one student.

Interview Questions

Which layer does hub work on?

Physical Layer.

Why is hub slow?

Because it broadcasts data to all devices.

Why collisions occur in hub?

All devices share same bandwidth.

2. SWITCH

Technical Definition

Network Switch

Switch is a networking device that forwards data only to the intended destination using MAC addresses.

Works on:

- Data Link Layer (Layer 2)
-

Why Switch was Introduced?

Hub created many problems:

- Collisions
- Slow communication
- Poor performance

Switch solves these issues.

MAC Address

Technical Definition

MAC Address

MAC address is a unique physical address assigned to every network interface card.

Example:

00:1A:2B:3C:4D:5E

Working of Switch

Switch maintains a MAC Address Table.

When a frame arrives:

1. Switch reads source MAC
2. Stores it in MAC table
3. Reads destination MAC
4. Sends frame only to correct device

Unlike hub, switch does NOT broadcast every frame.

Example

If PC1 sends data to PC3:

- Only PC3 receives data

Other systems do not receive it.

MAC Address Table

MAC Address Port

A1 Port 1

MAC Address Port

B1 Port 2

This helps switch forward frames efficiently.

Collision Domain in Switch

Each switch port forms separate collision domain.

Therefore:

- Very few collisions
 - Better performance
-

Broadcast Domain

Entire switch network remains same broadcast domain unless VLAN is used.

VLAN

Technical Definition

Virtual LAN

VLAN divides one physical network into multiple logical networks.

Purpose:

- Better security
 - Better traffic management
-

Characteristics of Switch

Feature	Description
Layer	Data Link
Address Used	MAC
Communication	Intelligent forwarding
Collision	Very low
Duplex	Full duplex

Advantages of Switch

- Faster communication
 - Better security
 - Reduced collisions
 - Efficient bandwidth usage
-

Disadvantages of Switch

- More expensive than hub
 - Requires management in large networks
-

Real Life Analogy

Courier delivering parcel only to correct address.

Interview Questions

Why switch is faster than hub?

Because switch sends data only to intended device.

Which address switch uses?

MAC address.

Which layer switch works on?

Data Link Layer.

3. ROUTER

Technical Definition

Router

Router is a networking device that connects multiple networks and forwards packets using IP addresses.

Works on:

- Network Layer (Layer 3)
-

Main Purpose of Router

Router connects different networks together.

Example:

- Home network ↔ Internet

IP Address

Technical Definition

IP Address

IP address is a logical address used to identify devices across networks.

Example:

192.168.1.1

Working of Router

When packet arrives:

1. Router reads destination IP
2. Checks routing table
3. Selects best path
4. Forwards packet

Routing Table

Stores path information.

Destination Next Hop

192.168.1.0 Port 1

Routing

Technical Definition

Routing

Routing is the process of selecting best path for packets.

Types of Routing

1. Static Routing

Manually configured routes.

2. Dynamic Routing

Routes updated automatically.

Uses protocols like:

- RIP
- OSPF
- BGP

Characteristics of Router

Feature	Description
Layer	Network
Address Used	IP
Function	Routing
Intelligence	High

Advantages of Router

- Connects different networks
- Internet sharing
- Better security
- Traffic control

Disadvantages of Router

- Expensive
- Complex configuration

Real Life Analogy

Google Maps choosing best road.

Interview Questions

Difference between switch and router?

Switch	Router
Uses MAC	Uses IP
Same network	Different networks

Which layer router works on?

Network Layer.

Why router needed?

To connect different networks.

4. REPEATER

Technical Definition

Repeater

Repeater is a device that regenerates weak signals to extend communication distance.

Works on:

- Physical Layer
-

Why Repeater Needed?

Signals become weak over long distances.

Repeater:

- Receives weak signal
 - Amplifies/regenerates it
 - Sends stronger signal
-

Advantages

- Extends network range
 - Improves signal quality
-

Disadvantages

- Cannot filter traffic
-

5. BRIDGE

Technical Definition

Network Bridge

Bridge connects two LAN segments and filters traffic using MAC addresses.

Purpose of Bridge

Reduce unnecessary traffic.

Improves network efficiency.

Difference Between Bridge and Switch

Bridge	Switch
Few ports	Many ports
Less intelligent	More intelligent

6. MODEM

Technical Definition

Modem

Modem converts digital signals into analog signals and vice versa.

Full Form

MODulator + DEModulator

Why Modem Needed?

Internet lines often use analog signals while computers use digital signals.

Modem performs conversion.

7. FIREWALL

Technical Definition

Firewall

Firewall monitors and filters incoming and outgoing traffic based on security rules.

Purpose

Protect network from:

- Hackers
 - Malware
 - Unauthorized access
-

Real Life Analogy

Security guard checking entries.

8. ACCESS POINT

Technical Definition

Wireless Access Point

Access Point allows wireless devices to connect to wired network using WiFi.

Purpose

Provides wireless connectivity.

Example:

Office WiFi.

9. NIC (Network Interface Card)

Technical Definition

Network Interface Card

NIC is a hardware component that allows a computer to connect to a network.

It provides:

- MAC address

- Network communication capability
-

Types

- Wired NIC
 - Wireless NIC
-

Final Important Differences

Hub vs Switch vs Router

Feature	Hub	Switch	Router
Layer	Physical	Data Link	Network
Address Used	None	MAC	IP
Communication	Broadcast	Intelligent forwarding	Routing
Speed	Slow	Fast	Fast
Intelligence	No	Moderate	High

Most Important Interview Questions

1. Difference between hub and switch?
2. Difference between switch and router?
3. Which device uses MAC address?
4. Which device uses IP address?
5. Which layer does router work on?
6. What is collision domain?
7. What is broadcast domain?
8. Why switch better than hub?
9. What is routing?
10. Why repeater needed?